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To cite this article: Dan M Kluger *et al* 2025 *Environ. Res. Lett.* **20** 114037

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RECEIVED
9 June 2025

REVISED
26 September 2025

ACCEPTED FOR PUBLICATION
3 October 2025

PUBLISHED
17 October 2025

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LETTER

Precrop payoffs: causal machine learning reveals large but variable yield benefits of crop rotation in major breadbaskets

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Keywords: crop rotation, precrop effects, remote sensing, causal inference, heterogeneous treatment effects, rotational diversity

Supplementary material for this article is available online

Abstract

Building sustainable food systems that are resilient to climate change will require improved agricultural management and policy. One common practice that is well-known to benefit crop yields is crop rotation, yet there remains limited understanding of how the benefits of crop rotation vary for different crop sequences and for different weather conditions. To address these gaps, we leverage crop type maps, satellite data, and causal machine learning to study how precrop effects on subsequent yields vary with cropping sequence choice and weather. Complementing and going beyond what is known from randomized field trials, we find that (i) for those farmers who do rotate, the most common precrop choices tend to be among the most beneficial, (ii) the effects of switching from a simple rotation (which alternates between two crops) to a more diverse rotation were typically small and sometimes even negative, (iii) precrop effects tended to be greater under rainier conditions, (iv) precrop effects were greater under warmer conditions for soybean yields but not for other crops, and (v) legume precrops conferred smaller benefits under warmer conditions. Our results and the methods we use can enable farmers and policy makers to identify which rotations will be most effective at improving crop yields in a changing climate.

1. Introduction

Crop rotation is a well-established agronomic practice that typically raises crop yields while reducing the need for fertilizer and pesticide inputs. Yield benefits can arise from several mechanisms, including suppression of harmful weeds, pests, and diseases [1, 2], increase of soil availability of essential nutrients such as nitrogen [3–5], and improved soil health and microbial biomass [6, 7]. Although these mechanisms are well established, less is known about how the benefits of crop rotation depend on the specific cropping sequence and on the weather conditions. This knowledge gap makes it difficult to assess how sub-optimal current farming practices are (in many regions farmers often grow the same crop each year or use a simple 2-crop rotation) and whether shifts in crop rotation could help farmers adapt to ongoing climate changes. Answering these questions is crucial but will require moving beyond traditional research approaches that leverage data from multiple randomized field experiments.

Randomized field experiments are a highly reliable approach for assessing the causal effects of crop rotation. However, due to the limited number of experiments with publicly available yield data and the wide variety of management practices, study designs and crop rotations used in these experiments, it is difficult to study how the benefit of any particular type of crop rotation varies with climatic conditions using experimental data alone. Studies that make claims about how rotation benefit varies with weather either use a limited number of experiments [8] or aggregate the analysis across multiple precrops

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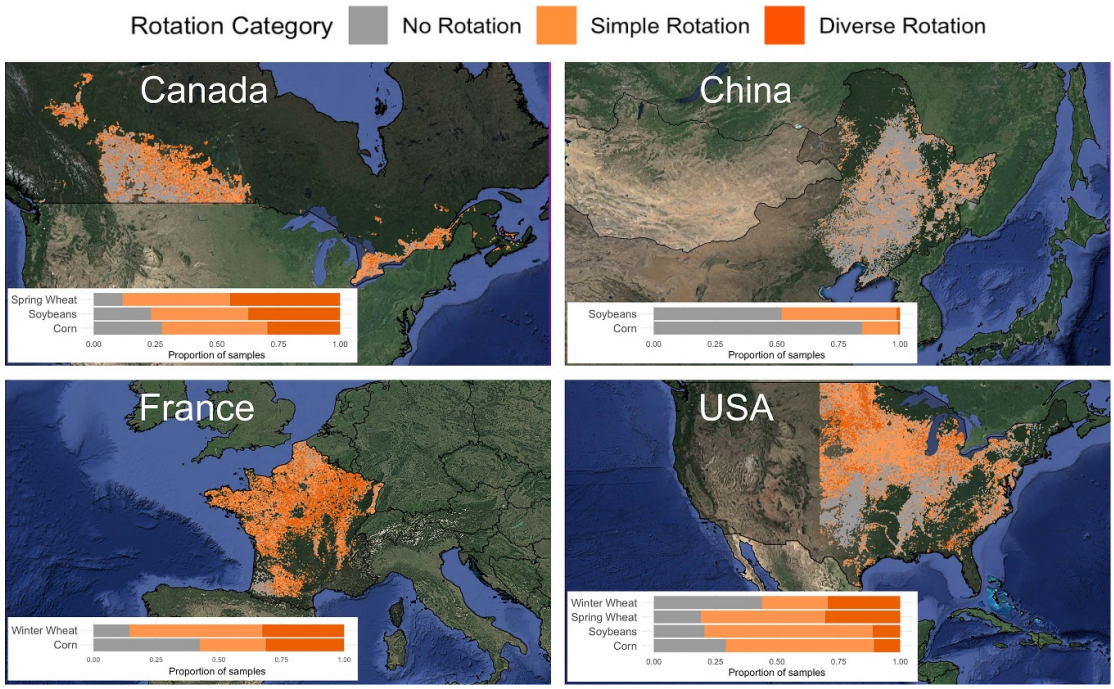

Figure 1. Rotation frequencies in study regions for a typical year. The maps for each country only show points in which one of the outcome crop types of interest was grown in 2019, and uses the crop type history from 2017 to 2019 to determine the rotation category. Here, the rotation categories are no rotation (i.e. with a 2 year sequence of the form B → B), simple rotation (i.e. any 3 year sequence of the form B → A → B or A → A → B where crop A is different than crop B), and diverse rotation (i.e. any 3 year sequence of three distinct crops). For each country and outcome crop of interest, the barcharts give the proportion of all samples (across all years, not just 2019) in our study that had each type of rotation. Note that the proportions displayed in the barcharts are not precise estimates of the prevalence of rotation, because they are calculated from crop type maps with classification errors, and the samples were taken to over-represent rotations involving crops that we considered studying as outcome crops (section A.1 and appendix D).

or outcome crops [8, 9]. Moreover, metanalyses that study the effects of diversifying crop rotation sequences (by adding at least one additional crop type to the sequence) typically have a different set of simple control rotations and diverse treatment rotations for each experiment in the study [5, 10, 11]. These studies use metrics of rotational diversity and fit a model to estimate the benefit of diversifying crop rotations without considering the particular crop type sequence. To our knowledge, how the benefit of diversification varies with precrop and outcome crop choices has not been studied and is a limitation noted in [11].

Recent advances in crop type mapping, satellite imagery, and causal machine learning have enabled researchers to assess the benefit of crop rotations using much larger sample sizes which allow examination of how rotation benefits vary with weather, rotational diversity, and the specific precrop and outcome crop. A number of recent studies used crop type maps and satellite imagery to assess the benefit of crop rotations [12–21], and the accuracy of some of these observational approaches have been validated against data from randomized field experiments [12] or expert recommendations [18]. However, all of these studies have limitations in terms of their scope, their methodology, or both (see table S6 for details). In summary, none of these studies assess the impact of switching from a simple rotation to a diverse rotation, and only a few of them assess how the benefit of rotation varies with weather [12, 17, 19, 20]. Moreover, most of these studies do not use a formal causal inference framework, and instead often rely on simple comparisons between rotated and non-rotated samples. Only [12, 19, 20] use a formal causal inference framework by leveraging modern causal machine learning tools. Of these studies, only [20] quantifies the uncertainty in their estimates of how the benefit of rotation varies with weather. As with the current study, most of these studies focus on precrop effects (defined as the effect of the previous year's crop type on the yield of the current crop) rather than total or long-term yield effects of rotation on all crops in the sequence. The former is a major (but not the only) component of rotation effects, while the latter are difficult to detect given the limited number of years in which large-scale, high-resolution crop type maps are available.

In this paper, we leverage crop type maps, satellite imagery, and a recently developed causal machine learning tool [22] to study the precrop effects and benefits of diversification in four different countries where crop rotation is common but not universally adopted (figure 1). For each outcome crop and commonly used precrop, we estimate the precrop effect and the impact of switching from a simple

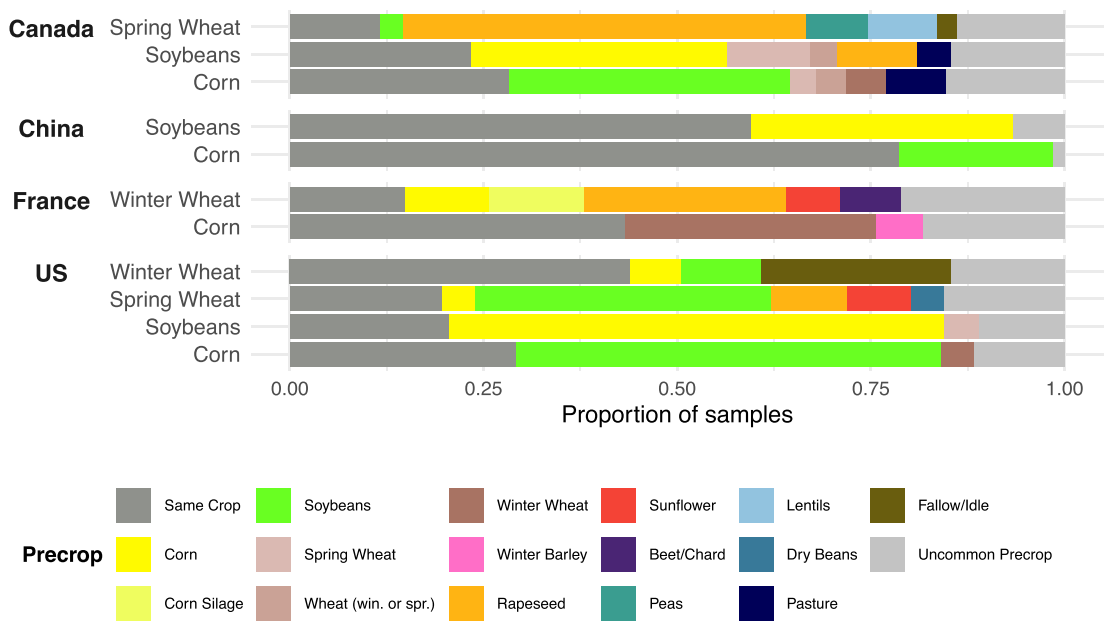

Figure 2. Distribution of precrops in our sample. Each sample corresponds to a unique pixel and year pair and was drawn according to the sampling scheme described in section A.1 and appendix D (pixels were of size $\leqslant 30\,\mathrm{m} \times 30\,\mathrm{m}$). For each country and outcome crop of interest, we plot the distribution of the precrop type among our samples from that country for which that outcome crop was growing in the corresponding year. The dark gray represents the proportion of samples where the precrop and the outcome crop are the same crop type. The light gray represents the proportion of samples that were excluded from our study due to an uncommon precrop (section A.3). We remark that the proportions displayed here are not precise estimates of the prevalence of rotation. In particular, the samples were taken in such a way that overrepresents rotations involving crops that we considered studying as outcome crops (section A.1 and appendix D). In addition, in Canada, China and the US, the presented proportions are inferred using satellite-based crop type maps with classification errors (e.g. as noted in [23] barley and wheat are difficult to distinguish with satellite imagery which can render barley underrepresented in the figure).

cropping sequence to a more diverse sequence. We also study how the precrop effects vary with weather and provide uncertainty quantifications for these estimates. Within each country, our study focuses on major outcome crops (among corn, soybean, winter wheat and spring wheat) and the most prevalent precrops used for these outcome crops (figure 2). In addition, our methodology can readily be deployed in any country with annual crop type maps or to study the benefits of rotations involving other precrop and outcome crop combinations not investigated here.

2. Overview of methods

This section gives an overview of the data and methods. For a more detailed description of the data and methods used and for explanations of some of the concepts and choices in the analysis, see the supplementary materials and methods section.

2.1. Dataset and rotations studied

We used annual crop type maps from northeastern China [24] (2017–2019), Canada [25] (2011–2020), France [26] (2015–2021), and the United States [27] (2008–2021). Using these crop maps and Google Earth Engine, we extracted a random sample of geographical points among croplands of interest. At each sampled location, monthly precipitation, monthly maximum and minimum temperature averages, and monthly vapor pressure deficit (VPD) were extracted from the TerraClimate gridded weather dataset [28]. For each sampled location and year, we extracted a time series of various spectral and quality assessment bands from Sentinel-2 (in China and France) and Landsat (in Canada and the US). For more details on the data sources, how they were harmonized across multiple spatial resolutions, and the sampling scheme, see section A.1.

For each sampled location and year, the time series of multispectral satellite data was converted to a normalized measure of peak greenness, which serves as a proxy for crop yield (see section A.2 for details). In summary, we first computed a time series of the green chlorophyll vegetation index (GCVI) [29]. We then estimated the peak GCVI in each location and year by removing observations that were corrupted by cloud cover (according to the satellites' quality assessment bands), fitting a harmonic regression, and calculating the peak of the best fit curve. GCVI was originally designed to capture chlorophyll content in crops [29] and peak GCVI has been shown to be a useful predictor of crop yields [30–33]. For increased interpretability, estimated peak GCVI values were normalized to have mean 1 for each crop type in each country.

For each location and year in our sample, we defined the precrop to be the crop grown in the previous year and the outcome crop to be the crop grown

in the current year. Samples where the precrop and outcome crop were the same were classified as having no rotation and were used as control units. Samples where the outcome crop B and precrop A were different were considered treated units and were categorized as having an A → B cropping sequence. In each country, we restricted our attention to at most four outcome crops of interest and for each outcome crop and country we studied the impact of the most prevalent precrops in the sample (figure 2). See section A.3 for more details about the precrop and outcome crop selection criteria used.

2.2. Estimating precrop and diversification effects

We fit causal forests [22, 34] to estimate the precrop effects. A causal forest is a recently developed method for conducting causal inference in observational studies, and it can be used to estimate the average treatment effect as well as the treatment effect as a function of the covariates. Causal forests have been used to study the effects of various agronomic practices such as tillage [35], cover cropping [36], and crop rotation [12, 20] on crop yields as well as the impact of crop rotation on net primary productivity [19]. In the context of assessing the effect of the Soybean → Corn crop rotation on corn yield, rotation benefits estimated by fitting a causal forest to a satellite-derived dataset were found to have a statistically significant positive correlation with estimates from actual randomized field experiments [12].

For each of four countries, and for each common combination of precrop A and outcome crop B, we conducted the following procedure to estimate the impact of the A → B sequence compared to the B → B sequence on a proxy for the yield of B in the second year. First, we subsetted our data to only include samples with A → B or B → B sequences and designated a binary treatment variable Z to be 1 for samples with A → B sequences and 0 for samples with B → B sequences. Next, we set the outcome variable V to be the normalized peak GCVI variable described in section A.2, which serves as a proxy for crop yield. We then let X denote a vector of year, latitude, longitude, 7 weather covariates, and (in the US only) irrigation status (from the landsat-based irrigation dataset [37]). The weather covariates captured early precipitation, growing season precipitation, average maximum and minimum daily temperatures in the growing season, and VPD for the three peak months of the growing season. We then fit a causal forest, using the `grf` package in R [38], to estimate the function

$$\omega(x) = \mathbb{E}[V|X = x, Z = 1] - \mathbb{E}[V|X = x, Z = 0].$$

In words, $\omega(\cdot)$ is the difference in the mean normalized peak GCVI in the treated group minus that in the control group written as a function of the covariates x. Under the assumption that there are no unmeasured confounders, $\omega(x)$ is equal to the conditional average treatment effect (which is the mean causal effect of the treatment Z on the outcome V as a function of x). Finally the average precrop effects were estimated by taking an overlap-weighted average of the estimated $\omega(X)$ values, assigning little weight to locations in space and time where only $Z = 1$ or only $Z = 0$ was observed and $\omega(X)$ could not have been properly estimated.

We refer readers to section A.4 for a more explicit description of the causal forest approach, the parameters and software used, the estimand, how standard errors were calculated, and the specific weather covariates used. That section describes various justifications and robustness checks regarding the choice of outcome variable, the choice of an overlap-weighted estimand, and our choice to not account for spatial correlations when computing standard errors (see also appendix C.5). We also compared some of the causal forest results to those from a simpler approach of calculating within each $1^\circ \times 1^\circ$ grid cell the difference between the mean of V in the treated (rotated) versus control (non-rotated) group, finding similar qualitative results (see appendix C.6). For model fit accuracy metrics, see appendix K and table S8.

To estimate the impact of diverse crop rotations compared to simple rotations, we used the same causal forest methodology with a different choice of treatment variable Z that was based on the three year rotation history. Within each country (except China), and for each combination of precrop A and outcome crop B, samples where a crop other than A or B was grown before A were considered to have a diversified rotation and were deemed as treated units ($Z = 1$) whereas samples where B → A → B was the three year rotation history were considered as the simple rotation control units ($Z = 0$). See section A.6 for more details.

2.3. Estimating heterogeneity of precrop effects with weather

To assess how the benefit of the A → B sequence versus the control B → B varied with weather, we ran the following linear regression of $\omega(X)$ on the year t, the growing season precipitation P, and the growing season temperature T

$$\omega(X) = \beta_0 + \beta_{\text{Year}} t + \beta_{\text{Precip}} P + \beta_{\text{Temp}} T + \varepsilon.$$

The above regression was implemented using the `best_linear_projection` function in the `grf` package [38], which provided standard errors and p-values that appropriately accounted for the fact that the function $\omega(\cdot)$ was unknown and had to be estimated from the data. The regression was fit on the subsample of points where the two year sequence was either A → B or B → B and excluded regions where the estimated propensity score was between 0.05 and 0.95 (allowing us to ignore regions where only A → B

or only B → B was observed, but not both). For more details about the subsample used, the growing season weather variables, how confidence intervals were constructed, and for a justification of the linear modeling choices see section A.5. As a supplementary sensitivity analysis, variants of the above regression model that included geographical fixed effects were also considered (appendix C.4).

To rescale the estimated regression coefficients to be on a more interpretable scale, on the same subsample, we computed the 25th percentile and 75th percentile for growing season precipitation and for growing season temperatures denoted by $P_{0.25}$, $P_{0.75}$, $T_{0.25}$, and $T_{0.75}$, respectively. Letting $\hat{\beta}_{\text{Precip}}$ and $\hat{\beta}_{\text{Temp}}$ denote the estimated regression coefficients from the above regression, our rescaled heterogeneity coefficients for temperature and weather were calculated with

$$\hat{\beta}^{(\text{resc.})}_{\text{Precip}} = \hat{\beta}_{\text{Precip}} \times (P_{0.75} - P_{0.25}) \quad \text{and}$$
$$\hat{\beta}^{(\text{resc.})}_{\text{Temp}} = \hat{\beta}_{\text{Temp}} \times (T_{0.75} - T_{0.25}).$$

The rescaled heterogeneity coefficients can be roughly interpreted as the amount by which the estimated benefit of rotation on our normalized yield proxy are expected to change as the growing season precipitation (or temperature) increases from the 25th percentile to the 75th percentile in the regions where both the rotation A → B and the no rotation control B → B are commonly observed. The above process was repeated for each precrop A, outcome crop B, and country listed in table S9.

2.4. Calibration with crop yield data

To improve the interpretability of our results, we converted all estimated precrop effects, diversification effects, and heterogeneity coefficients from a unitless GCVI scale to the crop yield scale by using a post-hoc, linear calibration with subnational-level crop yield data. In particular, we fit linear regressions of annual county-level crop yield data from [39] on county-level averages of normalized peak GCVI. Best fit linear regression coefficients were then used to convert all results to be in units of % of the average yield for the outcome crop and country of interest. For more details about the calibration and sensitivity analyses with alternative calibration schemes see section A.7 and appendix J.

3. Results

3.1. Estimates of impacts of 2 year crop sequences

The results (table S9 (column 6) and figure 3) indicate that rotations generally benefit crop yields, but there are considerable differences in these benefits for different precrop and outcome crop combinations. Of the four outcome crops considered, the rotation benefit estimates were generally largest for spring wheat and smallest for corn. For winter wheat outcome crops, soybean precrops had a negative estimated effect whereas rapeseed or fallow precrops had a positive estimated effect. For spring wheat outcome crops, legume precrops and rapeseed precrops were generally the most effective, whereas for soybean outcome crops, cereals were the most effective types of precrops (figure S1).

The results also suggest that beneficial precrops are being exploited by farmers. In figure 3, we see that the most common precrops have either the highest or nearly the highest benefits. There are some exceptions in which less common precrop choices are estimated to be more effective than the most common precrop choices when spring wheat or corn is the outcome crop. In addition, only relatively common precrops are included in the analysis (section A.3), so our analysis does not rule out the possibility that some very rare precrops are more effective than commonly used precrops.

3.2. Variability of rotation benefits with weather

The results (table S9 (columns 8–9) and figure 4) indicate that at higher growing season precipitation there is generally a greater benefit due to rotation. In particular, with a few exceptions in France and for fallow precrops, cases where the rotation benefit is greater at lower precipitation levels did not have statistically significant heterogeneity coefficients. In addition, based on a sensitivity analysis where we use the log of estimated crop yield as the outcome variable (appendix C.3 and figure S5, bottom left panel), we find that this result is not merely driven by the fact that crop productivity is higher in rainier conditions. Instead, we see increases in both the relative effectiveness and absolute effectiveness of crop rotation in rainier conditions. For the rotations with a legume or broadleaf precrop, the heterogeneity of the rotation benefit with precipitation was larger and more consistently statistically significant than it was for the rotations where the precrop was a cereal. These findings are generally robust to sensitivity analyses where we include fixed effects to control for geographical subregions when estimating the heterogeneity of the rotation benefits with weather (appendix C.4 and figure S6), although for cereal precrops our sensitivity analyses did not clearly suggest that the rotation benefit was higher at higher precipitation.

The direction of how the rotation benefit varies with temperature was not consistently positive or negative; however, some notable patterns exist. When soybean is the outcome crop (red bars in figure 4), the benefit of rotation is generally greater at higher growing season temperatures. In addition, when there is a legume precrop, the benefit of rotation is generally smaller at higher growing season temperatures (top right panel in figure 4). The only exceptions

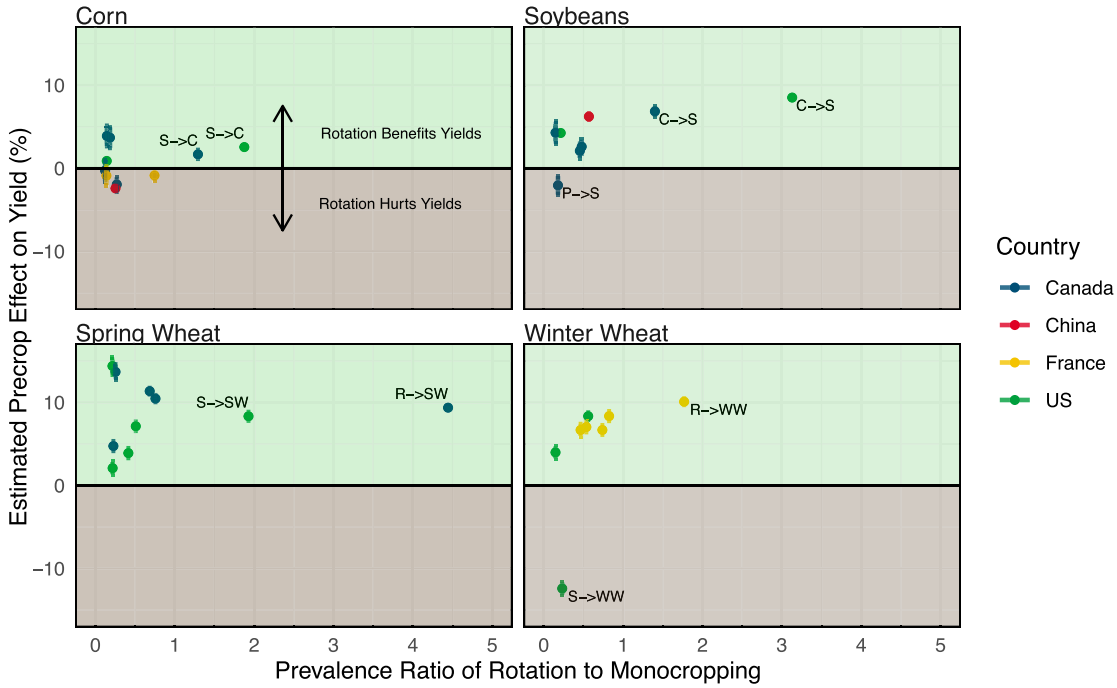

Figure 3. Estimated precrop effects. Each point represents a different country, precrop, and outcome crop combination, with the panel titles giving the outcome crop and the colors giving the country. For each point, the *y*-axis value gives the estimated average effect of a 2 year cropping sequence on crop yield (expressed as a percentage of the mean crop yield of the corresponding outcome crop and study region) and the error bars give 95% confidence intervals. The estimates and standard errors were computed using the approach described in section A.4 and were converted to the yield scale using the approach described in section A.7. The confidence intervals are based on large samples and are much narrower than one might expect because they cannot reflect uncertainty in the potential bias due to unmeasured confounders. The *x*-axis gives the prevalence ratio, which we define as the number of samples that were used as treated units and had a given 2 year cropping sequence divided by the number of samples that were used as control units where the precrop and outcome crop matched. The specific type of rotation is labeled in cases where the prevalence ratio was greater than 1 and in cases where the rotation had a uniquely negative estimated effect. For point labels, S = Soybean, C = Corn, P = Pasture and Forages, R = Rapeseed, SW = Spring Wheat, WW = Winter Wheat.

are cases where the heterogeneity coefficient is not statistically significant. Moreover, sensitivity analyses (appendix C.4, figure S7) demonstrate that these findings are robust to the addition of various geographical fixed effects in our model used to assess the heterogeneity of the rotation benefits.

3.3. Estimated impacts of diversified crop rotations

The results suggest that diversification beyond a simple rotation confers large additional benefits in some cases, but more often the benefits are small, not statistically significant, or even negative (table S9 (column 7) and figure 5). Spring wheat appears to benefit most consistently from more diverse rotations, whereas when corn is the outcome crop, the estimated impact of diversification is sometimes negative, particularly for common precrop choices (soybean in Canada and the US, and winter wheat in France). Supplementary results also showed that the impact of diversification beyond a simple rotation tended to be larger in rainier conditions (figure S2), although many estimates of the heterogeneity of the diversification effects were not statistically significant due to limited sample sizes in the diversification analyses.

4. Discussion

4.1. Discussion of rotation benefit estimate findings

Our results are consistent with the well established principle that switching from no rotation to crop rotation generally benefits the subsequent crop's yields, providing some validation to our approach. For corn yields, our results suggest that winter wheat and soybean were beneficial precrops in the United States and Canada, although this conclusion could not be drawn in China and France or for spring wheat precrops. For soybean yields, corn was a particularly beneficial precrop (in the US, Canada, and China), while wheat (in the US and Canada) and rapeseed (in Canada) precrops were found to have smaller, yet still positive benefits on yields. For spring wheat yields in the US and Canada, rapeseed and legumes such as soybeans, peas, lentils and drybeans were found to be particularly beneficial precrops, whereas in the US sunflower and corn precrops had more modest benefits. For winter wheat yields in France, rapeseed was again found to be a particularly effective precrop, and sunflower, and chard/beet, corn silage, and corn, were also quite effective as precrops, with the latter finding for corn precrops also holding in the US. Overall,

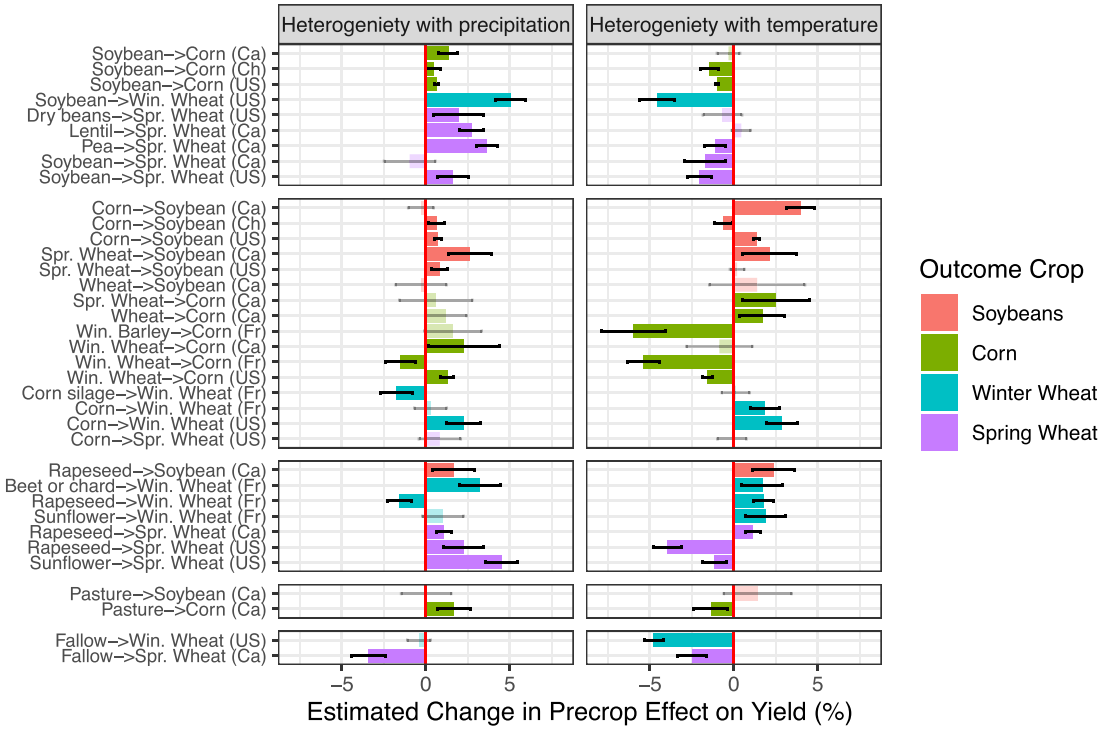

Figure 4. Estimates of the heterogeneity of rotation benefits with weather. Each row gives the two year crop sequence, the country from which the study sample was taken (Ca = Canada, Ch = China, Fr = France, US = United States), and the rescaled regression coefficients for growing season precipitation (left column) and temperature (right column) which estimate how much the precrop effect changes when the precipitation or temperature increases from the 25th percentile to the 75th percentile of observed values. The x-axis values are in units of percent of the mean crop yield of the corresponding outcome crop in the corresponding study region and can be converted to units of tons per hectare using the yield estimates in table S10. The error bars give 95% confidence intervals (based on heteroskedasticity-robust (HC3) estimation), and coefficient estimates that are not statistically significant (level $\alpha = 0.05$ based on two-sided testing) are given a lighter, translucent color. The rows are grouped into blocks based on 5 precrop categories of interest: legumes, cereals, broadleaf crops, pasture/forages, and fallow.

the precrop effects that we estimated were generally positive and statistically significant, with two exceptions when corn was the outcome crop and soybean or wheat was the precrop, two exceptions when pasture was the precrop, and a notable exception of the Soybean → Winter Wheat sequence in the US.

The negative effects for corn can be partly explained by reduced fertilizer use in rotations. For example, in the US, farmers growing Soybean → Corn are advised to use less fertilizer than those growing Corn → Corn [40]. Prior work [12] found that estimates of the impact of the Soybean → Corn rotation using a similar observational approach that omitted fertilizer use underestimated the rotation benefit estimates based on randomized field experiments (in the experiments, fertilizer use was typically the same in rotated and non-rotated subplots). Omission of fertilizer also explains why the estimated precrop effects of soybean on corn were much smaller than the estimated precrop effects of corn on soybean, contrary to findings from randomized field experiments. Negative effects in Canada for cases where pasture was a precrop could be explained by the fact that Pasture → Soybean and Pasture → Corn samples often reflected a conversion of pastoral land to cropland rather than a true rotation, and thus may correspond to less productive land. Indeed, among our samples from Canada that had either a Pasture → Corn or a Pasture → Soybean sequence, roughly 64% observed a switch from a pastoral classification to a nonpastoral classification only once between 2011 and 2020, and 29% observed such a switch twice. Further, when using a different satellite-based proxy for yield in our sensitivity analysis in appendix C.2, the negative estimated effects of the Pasture → Corn and Pasture → Soybean rotations were no longer statistically significant.

We estimated a large negative effect of the Soybean → Winter Wheat rotation on subsequent winter wheat yields in the United States. One possible explanation for this is that winter wheat has higher yields at earlier planting dates [41], but the Soybean → Winter Wheat rotation can prevent early planting of winter wheat. According to a USDA report [42], the typical winter wheat planting window started and ended earlier than the typical soybean harvest window in most of the states that were highly represented in our study's Soybean → Winter Wheat and Winter Wheat → Winter Wheat samples (figure S12). In addition, some experiments have suggested that soybean precrops lead to reduced soil water content

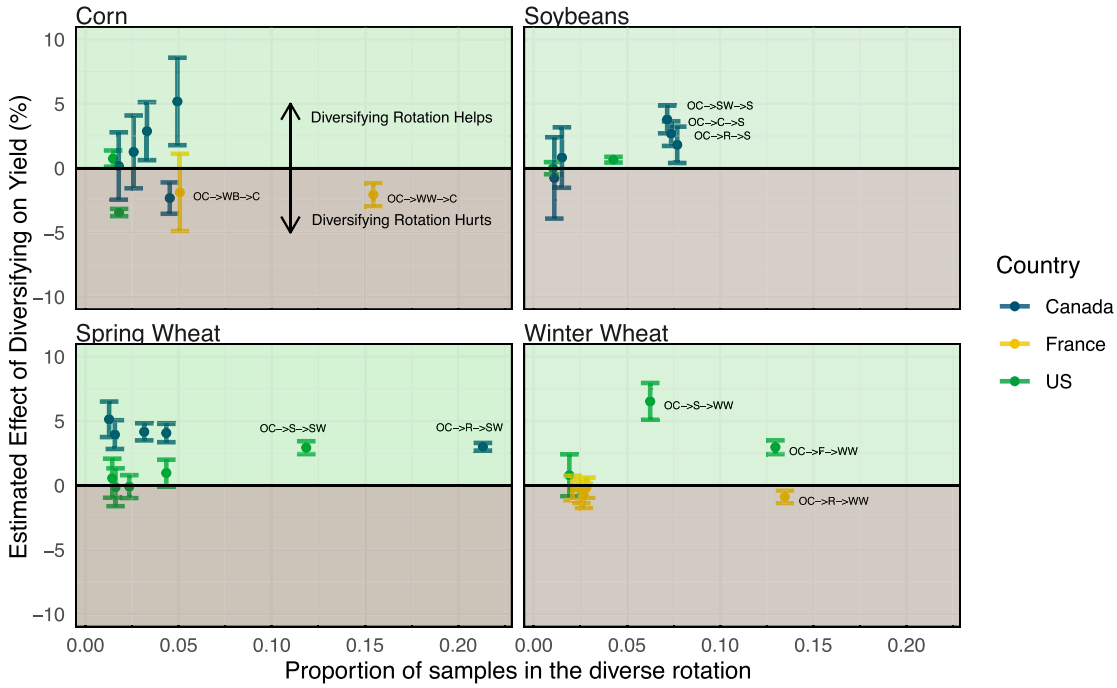

Figure 5. Estimated benefits of diversifying crop rotations. Each point represents a different country, precrop, and outcome crop combination, with the panel titles giving the outcome crop and the colors giving the country. For each point, the *y*-axis value gives the estimated effect of diversifying the rotation (by growing a crop other than the precrop or outcome crop in the year prior to the precrop) on crop yield, and the error bars give 95% confidence intervals. The estimates and standard errors were computed using the approach described in sections A.4 and A.6 and were rescaled to be on the crop yield scale (expressed as a percentage of the mean crop yield of the corresponding outcome crop and study region) using the approach described in section A.7. The *x*-axis gives the proportion of samples that had a diversified rotation and a particular precrop, among all samples with the given outcome crop in the given country. The specific type of diversified rotation is labeled in cases where this proportion exceeded 0.05. For point labels, S = Soybean, C = Corn, SW = Spring Wheat, WW = Winter Wheat, WB = Winter Barley, F = Fallow, R = Rapeseed, and OC = Other Crop which groups together *all* other crops besides the specified precrop and outcome crop.

in the subsequent spring because their residues are ineffective at trapping snow compared to the residues of cereals such as corn and sorghum [43]. Meanwhile, residues from winter wheat (typically harvested in the summer) are effective at trapping snow (in the subsequent winter), particularly when the residues are left standing [44]. Therefore, in semi-arid regions with snow and in fields that use favorable residue management practices, a winter wheat precrop leads to more soil water availability in the subsequent spring than a soybean precrop does. Low yields in the Soybean → Winter Wheat rotation have also been reported in the experimental literature, although the results are mixed. A field experiment in the north China plain found that the Soybean → Winter Wheat rotation led to lower Winter Wheat yields than in a Corn → Winter Wheat rotation, particularly in dry years [45] which the authors attributed to soybean reducing the soil available water for subsequent wheat growth. Contrary to our findings, an experiment in Woodslee, Ontario found higher winter wheat yields in a Soybean-Winter Wheat rotation than in mono-cropped winter wheat [46]; however, the study was conducted in a location with high mean annual precipitation, and to our knowledge, the treated and control plots had identical planting dates. The fact that some Soybean → Winter Wheat versus Winter Wheat → Winter Wheat experiments use the same planting dates for the treatment and control subplots and are conducted in highly favorable weather conditions for the rotation serves as an example of how our observational approach can be used to uncover external validity issues in the experimental crop rotation literature.

We also find that some rotations are particularly beneficial for the outcome crop. For example, legumes appear particularly effective at improving subsequent spring wheat yields in the US and Canada (figure S1, table S9). This is consistent with field experiments in the region, which have shown that a variety of legumes, including soybeans and dry beans, increase nitrogen availability and yield of the subsequent spring wheat crop [47, 48]. A global metanalysis of randomized field experiments found legumes to be the most effective precrops for wheat [49], although the study did not distinguish between spring and winter wheat.

4.2. Discussion of heterogeneity with weather findings

Our results suggest that in rainier conditions, common crop rotations involving corn, soybean, and wheat generally have a greater impact on yields than in drier conditions. One likely reason is that higher precipitation increases pest and weed pressures, which crop rotations can help to address. In addition,

the residues of certain precrops, such as sunflower and soybean, are not effective at trapping precipitation from snow for subsequent soil absorption [43]. Therefore, assuming no residue removal, rotations with such precrops would be less effective when there is too little precipitation during the growing season to compensate for the reduction in soil moisture. Meanwhile, wheat stalks are particularly effective at trapping snow [44], so non-rotated wheat controls would perform relatively better under low levels of rain.

Increased benefits of rotation for soybean yields at higher temperatures likely relate to pest pressures [50–52], as these harmful pressures (that are mitigated by crop rotation) generally increase with warming. In contrast to soybean, we observe mixed effects of warming on rotation benefits for wheat and corn, and we find that rotations involving legume precrops are less effective at higher temperatures. The latter finding likely relates to the effect of warming on nitrogen limitations. At higher temperatures, residues of the precrop mineralizes at a faster rate, thereby reducing nitrogen limitations [53, 54]. The benefit from a legume precrop is therefore reduced with warming. Another potential explanation is that high temperatures can render soybean precrops as less effective in fixing nitrogen [55], and samples with high growing season temperatures are more likely to have had high temperatures while the soybean precrop was growing.

While these findings of how rotation effects vary with weather are consistent with the existing literature (appendix L), we note that our findings go far beyond the scope of most prior work. In particular, existing claims in the experimental literature looked at weather interactions after aggregating over many different precrop and outcome crop combinations, due to limitations in the number and types of experiments [5, 8–10]. Such claims can therefore reflect how precrops and outcome crops in experiments vary across regions with differing climates rather than how the benefit of rotation varies with weather. In addition, claims in the observational literature about how rotation benefits varied with weather all focused on a single country and only one of them provided uncertainty quantifications for these claims [12, 17, 19, 20] (table S6).

4.3. Discussion of diversification benefit findings

Our results suggest that diversification beyond a simple rotation has diminishing returns (see table S9 or figures 3 and 5). In particular, we find that the average benefit of a 2-crop rotation across our 36 rotation-country pairs is about 3.7 times larger than the average additional benefit of switching from 2-crop rotations to a more diverse sequence. This finding agrees with a recent metanalysis of 45 experiments in China [9] that found crop rotation increased yields by 20% when compared to monoculture on average but extended (diversified) rotations provided only an additional 4% yield boost. Similarly, a recent global metanalysis of 462 crop rotation experiments involving legumes found that the inclusion of an additional legume precrop in a rotation had lower benefits when the control rotation was already diverse [5]. A recent analysis of field experiments in the European Union and Africa [56] found no additional benefits from increased diversification beyond a switch from monocropping to simple rotations. A recent analysis of data from 32 long-term experiments from North America and Europe [11] found that crop yields for maize and small grain cereals increased at higher values of an index used to measure rotational diversity. However, when fitting quadratic regression models of crop yield on their diversity measure and other controls, the coefficient corresponding to the squared diversity measure was negative for all 3 outcome crop classes (maize, small grain winter cereal and small grain spring cereal) suggesting diminishing returns with each additional unit of diversity.

Due to experimental data limitations, recent metanalyses studying crop rotation diversification did not explore how the diversification benefit varied with each precrop and outcome crop combination [5, 9, 11, 56]. Our results suggest that the benefit of diversification varies considerably, and can even be negative (table S9 and figure 5). For example, we estimate a negative effect of diversification in settings where soybean is the precrop and corn is the outcome crop whereas we find considerable diversification benefits when there is a legume precrop and spring wheat is the outcome crop (table S9).

4.4. Limitations

This study has several limitations. First, our causal inference approach relies on the assumption that there are no unmeasured confounders. Due to data availability constraints we did not control for potential confounders such as soil quality (although, including soil variables in the US barley changed our results (appendix C.1)). We also did not account for management practices such as tillage, cover cropping, residue management, fertilizer use, and pesticide use; however, these unmeasured management practices can be viewed as unmeasured mediators as opposed to unmeasured confounders, and we estimate the net rather than direct effects of crop sequence choice (see section A.8). Moreover, farms that rotate tend to use less fertilizer [40] and pesticide [57], so our analysis does not capture the cost savings reductions in chemical inputs that are associated with rotations.

Second, we do not have exact measurements of the treatment and outcome variables which are instead estimated using satellite data. Misclassifications in the treatment variable likely lead to attenuation bias [12, 58], which as with the issue of unmeasured fertilizer and pesticide use, results in underestimation of rotation benefits. Appendix H suggests that our choice to use a vegetation index as

the outcome variable rather than crop yield estimates from [30, 59] does not substantially influence the results and we also find robustness to the choice of vegetation index (appendix C.2).

Third, the study has some limitations in its scope. We do not study the well-documented longer-term benefits of crop rotation [8, 11]. In addition, our analysis only considers the impact of a cropping sequence on the final crop in a cropping sequence rather than all crops in the sequence. Ultimately, farmers make decisions about which rotations to use based on many factors, including the expected yield and price of each of the crops in the rotation sequence; however, future work can couple our results and methods with crop rotation decision support systems [60–64].

A more comprehensive and detailed discussion of the limitations can be found in appendix B. That appendix also summarizes sensitivity analyses and mathematical results, which assess whether or not the limitations discussed are severe.

5. Conclusion

Using satellite data and causal machine learning, we estimated the impacts of various cropping sequences and how they varied with weather. While randomized field experiments can reliably estimate the effects of cropping sequences on crop yield, given the limited geographical span of experimental data, drawing conclusions of how rotation benefits vary with weather remains a substantial challenge. Our results using observational approaches suggest that precrop effects are larger under rainier conditions. Moreover, the results suggest that if the temperature increases, precrop effects will decrease in cases where the precrop is a legume but increase in cases where the outcome crop is soybean. Estimates of how rotation benefits vary with weather can be coupled with climate models to forecast how the effectiveness of crop rotation could change in a changing climate (as is done for the Soybean-Corn rotation in [20]) and can inform agricultural adaptation to climate change.

In addition, our observational approach can uncover or fill in gaps in the experimental crop rotation literature. For example, as discussed in section 4.1 our results for the Soybean → Winter Wheat sequence point to possible external validity issues regarding planting dates and weather conditions in the experimental literature. In addition, our approach can be used in any region with multiple years of crop type maps in order to identify cropping sequences that may be highly beneficial but have been understudied in randomized field experiments.

Data availability statement

Data and code that can be used to reproduce the analysis in this paper can be downloaded from the following URL/DOI: https://zenodo.org/records/15579276 [65].

Acknowledgments

This work was supported by a Stanford Interdisciplinary Graduate Fellowship and the NASA Harvest and ACRES Consortia (NASA Applied Sciences Grant No. 80NSSC23M0032 (sub-award 125062-Z6521205) and 80NSSC23M0034 (Sub-Award 124245-Z6512205)). The authors wish to thank Kevin Guo and Stefan Wager for helpful discussions and anonymous reviewers for helpful comments.

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